

Arunima Chakraborty

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EDUCATION

Christ University

Bachelor of Technology (Computer Science Engineering) – CGPA: 3.8/4

Bangalore, Karnataka

Aug. 2023 – May 2027

EXPERIENCE

AI & Robotics Research Intern

April 2026 – May 2026

National Institute of Technology (NIT) Agartala

Agartala, Tripura

- Developed a voice-controlled assistive robotic framework integrating Speech-to-Text (STT), Text-to-Speech (TTS), conversational AI, and Arduino-based embedded systems.
- Implemented intent recognition mechanisms to distinguish conversational inputs from robotic control commands, reducing false triggering during human-robot interaction.
- Established serial communication between Python applications and Arduino controllers for real-time voice-driven hardware automation.
- Conducted research on machine learning fundamentals, object detection, object tracking, assistive robotics, and indoor navigation systems for elderly and visually impaired users.

Cybersecurity Intern

April 2025 – May 2025

Directorate of Information

Agartala, Tripura

- Conducted basic website vulnerability assessment to identify flaws using cURL, Wireshark, and Burp Suite.
- Analyzed raw HTTP/HTTPS traffic, response codes, and intercepted secure requests for security insights.

PROJECTS

VGRS: Voice-Guided Room Scout | *Python, Arduino, Speech Recognition, Conversational AI*

2026

- Designed an assistive voice-guided navigation framework for elderly and visually impaired users, enabling natural human-robot interaction.
- Integrated Speech-to-Text (STT), Text-to-Speech (TTS), conversational AI, and embedded hardware communication for real-time assistance.
- Implemented intent classification techniques to differentiate conversational dialogue from actionable robotic commands.
- Proposed a scalable architecture supporting future integration with object detection, object tracking, obstacle awareness, and autonomous indoor navigation.

PersonaShield – AI-Enhanced Cybersecurity Analysis Platform | *Python, Flask, MongoDB, Cohere AI*

2026

- Designed and deployed a cybersecurity platform for detecting phishing, social engineering attacks, brand impersonation, and malicious URLs.
- Implemented AI-powered threat classification, explainable risk scoring, OpenPhish threat intelligence integration, and WHOIS-based domain reputation analysis.
- Built a Security Assistant capable of explaining threat indicators, attack techniques, and mitigation strategies using contextual AI responses.
- Created analytics dashboards, PDF threat reports, confidence scoring mechanisms, and MongoDB-backed threat intelligence storage for real-time security monitoring.

Resume Sudharak | *Python, Streamlit, Cohere API, LLM*

Oct 2025

- Architected an AI-powered resume analyzer using Streamlit and Cohere to quantify candidate-job fit via NLP matching.
- Implemented prompt engineering to generate personalized optimization feedback and automated skill gap analysis.

Lavasa: A Climate Blind Spot | *Data Visualization, Satellite Imagery*

Sep 2025

- Engineered an interactive dashboard utilizing satellite imagery to visualize ecological collapse and climate data gaps.
- Synthesized complex environmental datasets into actionable visual narratives to highlight urban planning blind spots.

Mentor-Mentee Connect | *React, Supabase, PostgreSQL, Auth*

Oct 2025

- Deployed a secure mentorship platform utilizing Supabase for Role-Based Access Control (RBAC) and real-time sync.
- Engineered robust CRUD APIs and authentication flows to efficiently manage session logging and user profiles.

FinSakhi | *Voice AI, Multilingual NLP, FinTech*

Dec 2025

- Built a voice-first multilingual financial assistant to enhance digital banking accessibility for diverse demographics.
- Achieved Finalist status at IEEE CS Geekathon 2025 for innovation in inclusive financial technology and NLP.

PUBLICATIONS

Enhancing Cybersecurity Risk Assessment through Machine Learning: A Study on Random Forest and XGBoost

- Co-authored a research paper exploring machine learning techniques for cybersecurity risk prediction using Random Forest and XGBoost models.
- Analyzed cybersecurity datasets to evaluate model accuracy and improve threat detection capabilities.

Detection of Cancer in the Early Stages of Childhood

- Co-authored a research paper focusing on early detection and diagnosis of childhood cancers through biological and clinical indicators.
- Reviewed existing medical literature and analyzed early-stage symptoms and diagnostic approaches to improve early intervention strategies.
- Explored biological risk factors and screening methods that support timely identification and treatment of pediatric cancers.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript, HTML, CSS

Databases: MongoDB (NoSQL), Oracle DB

AI & ML: Machine Learning Fundamentals, Speech Recognition, Conversational AI, Prompt Engineering, Object Detection, Object Tracking

Tools & Platforms: Arduino IDE, Streamlit, Git, Figma, Unity Hub, SolidWorks

Core Concepts: OOP, DSA, Embedded Systems, Human-Robot Interaction, Computer Vision

Soft Skills: Communication, Public Speaking, Analytical Thinking, Team Leadership

EXTRACURRICULAR ACTIVITIES

Volunteer – Education Cohort | Center for Social Action (CSA)

- Volunteered in the Education cohort, teaching specialized students and supporting inclusive learning initiatives.

Hub Volunteer | Google Cloud Agentic AI Day 2025

- Supported logistics and operations for 2400+ participants during a 30-hour innovation sprint hosted by Google Cloud and Hack2Skill.

Club Head | ASCII Club / CodeRush Club

- Led planning and execution of technical events, workshops, and coding challenges to foster peer learning and innovation.

Attendee | IISc Symposium on Data for Public Good

- Participated in the 2nd edition of the symposium at the Indian Institute of Science (IISc), Bangalore, exploring data-driven social impact.

Finalist | IEEE CS Geekathon 2025

- Developed FinSakhi, a voice-first multilingual financial chatbot empowering women through accessible digital banking.

Hackathon Participant | NASA Space Apps Challenge, HCL GUVI Hackathon

- Actively participated in national and international hackathons, contributing innovative solutions across domains such as space technology and education.